SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1703**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

*I **D.VAISHNAVI(192524143)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

ANSWER:

1(a) Citation, Domain and ML Problem

The selected article studies the use of **Deep Reinforcement Learning (DRL)** for recommending treatments to patients with **Type 2 Diabetes Mellitus (T2DM)**.

The main problem is that diabetes treatment is not a single decision. Patients may require a sequence of treatments over time, and different patients can respond differently to the same treatment. Therefore, selecting an appropriate treatment strategy is a **sequential decision-making problem**.

The researchers developed three treatment recommendation models for:

1. Antiglycemic treatment
2. Antihypertensive treatment
3. Lipid-lowering treatment

The system combined **clinical knowledge** with a **data-driven deep reinforcement learning model**. Clinical guidelines were first used to identify candidate medications, and the DRL model then ranked those candidates according to expected outcomes.

1(b) Key Objective and Research Gap

Objective

The main objective of the research was to develop personalized treatment recommendation models for patients with Type 2 Diabetes using Deep Reinforcement Learning. The authors wanted to address the difficulty of selecting appropriate treatments because diabetes patients have different health conditions, treatment histories, and responses to medication.

The study recognized that diabetes treatment is a **long-term sequential process**. A treatment decision made at one point can influence the patient's future health condition. Traditional approaches may not adequately capture these sequential effects. Reinforcement Learning is suitable because it learns actions according to their expected long-term rewards.

The researchers used historical clinical data to develop treatment recommendation models and then evaluated whether treatments consistent with the model were associated with better short-term and long-term outcomes. Three models were developed for antiglycemic, antihypertensive, and lipid-lowering treatments. The study therefore demonstrates how machine learning can support personalized healthcare decision-making while using clinical knowledge to constrain the possible treatment choices.

Research Gap

The article identifies an important gap: there can be a difference between **recommended treatment guidelines and the treatment actually received by patients**, while patients also have different therapeutic needs. The study addresses this gap by combining clinical knowledge with a data-driven reinforcement-learning approach.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

ANSWER:

2(a) ML Technique Used

The primary technique used was **Deep Reinforcement Learning**.

The approach consists of two major components:

1. Knowledge-driven model

Clinical guidelines and expert knowledge were used to identify appropriate candidate medications.

2. Data-driven model

A Deep Reinforcement Learning model was then used to rank the candidate treatments according to expected clinical outcomes.

This combination is useful because the model does not simply select arbitrary actions. Medical knowledge is incorporated before the data-driven model makes its recommendation.

Dataset

The study used data from the **Singapore Health Services Diabetes Registry**.

Dataset Information	Details
Patients	**189,520** patients with T2DM
Outpatient visits	**6,407,958**
Data period	**2013–2018**
Data source	Singapore Health Services Diabetes Registry
Training data	**80%**
Evaluation data	**20%**

Application	Type 2 Diabetes treatment recommendation

The test data contained **36,993 patients** for evaluating the effectiveness of the three treatment recommendation models.

The study considered clinical information and treatment histories to represent the patient's state and treatment decisions.

2(b) Mapping to CO4

CO4 Topic	Connection to Article
Inductive Learning	The model learns useful treatment policies from historical patient data.
Decision Trees	Not the primary technique used in this article.
Statistical Learning	Statistical methods were used in outcome evaluation and adjustment.
Reinforcement Learning	**Main technique** used to learn treatment recommendations.

Methodological Strength

A major strength is the combination of **clinical knowledge and data-driven reinforcement learning**. This makes the approach more appropriate for healthcare than relying only on an unrestricted data-driven model.

Methodological Limitation

A major limitation is that the study relies on **retrospective observational data** rather than a randomized clinical trial. Therefore, an association between model-concordant treatment and improved outcomes does not automatically prove that the model itself caused the improvement.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

ANSWER:

3(a) Key Results

The researchers evaluated the three treatment recommendation models using the remaining test data.

The test dataset contained **36,993 patients**. The researchers compared outcomes between treatments that were consistent with the model and treatments that were not consistent with the model. They used confounder-adjustment methods including:

- Stratification
- Propensity-score weighting
- Multivariate regression

For long-term outcomes, the researchers examined whether treatment-model concordance was associated with reduced occurrence of complications or death.

Comparative table

Aspect	Article Approach
Dataset	189,520 T2DM patients
Outpatient visits	6,407,958
Training	80%
Evaluation	20%
Test patients	36,993
Models	3 treatment recommendation models
Main technique	Deep Reinforcement Learning
Treatments	Antiglycemic, antihypertensive, lipid-lowering
Evaluation	Short-term and long-term outcomes

Important: The article is a clinical treatment-recommendation study rather than a simple classification problem. Therefore, it does **not** primarily report a conventional accuracy/F1/AUC comparison in the same way as a Decision Tree classification experiment. The appropriate evaluation is based on treatment concordance and clinical outcomes.

3(b) Critical Evaluation

Are the results realistic?

The large dataset makes the study reasonably strong from a data-volume perspective. The use of **189,520 patients and more than 6.4 million outpatient visits** provides substantial longitudinal information.

However, the results should not be interpreted as proof that the RL model can safely prescribe treatments independently.

Potential Biases

1. Selection bias

The data comes from a particular healthcare system and population. Results may not automatically generalize to every country or hospital.

2. Historical treatment bias

The model learns from treatments that were actually given by clinicians. Historical clinical decisions can contain existing biases or limitations.

3. Confounding

Patients receiving different treatments may differ in disease severity and other characteristics. The study attempted to address confounding using statistical adjustment, but observational data cannot completely eliminate this issue.

Additional experiments

The research could be strengthened by:

- Testing on an independent hospital dataset.
- Performing prospective clinical validation.
- Comparing with additional treatment-policy algorithms.
- Testing robustness across different demographic groups.
- Conducting longer-term external validation.

3(c) Real-World Applicability

The technique could potentially be used as a **clinical decision-support system** for personalized diabetes management.

Possible applications include:

1. Supporting doctors in selecting treatment options.
2. Personalizing treatment according to patient history.
3. Supporting long-term diabetes management.
4. Ranking candidate treatments according to predicted outcomes.

Industry Challenges

Data availability: Large, high-quality longitudinal patient datasets are required.

Data privacy: Medical records contain highly sensitive information and must be protected.

Model explainability: Doctors need to understand why a recommendation is produced.

Generalization: A model trained in one healthcare system may not perform equally well elsewhere.

Safety: Treatment recommendations require appropriate clinical validation and human oversight.

The broader literature also identifies challenges in defining states, actions, rewards, and safely deploying RL in healthcare.

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

ANSWER:

4(a) Three Key Insights

Insight 1 – Reinforcement Learning is suitable for sequential decisions

I learned that Reinforcement Learning is useful when decisions are made repeatedly over time and an action can influence future outcomes.

CO4 connection: Reinforcement Learning and sequential decision-making.

In diabetes management, a treatment decision at one visit can affect the patient's condition and future treatment decisions.

Insight 2 – State, Action and Reward are important

The article helped me understand how an RL problem can be represented using:

- **State:** Current patient health condition
- **Action:** Treatment option
- **Reward:** Clinical outcome

This directly connects the article with the **MDP concept** in CO4.

Insight 3 – Healthcare requires both AI and domain knowledge

I learned that machine learning should not necessarily operate independently in a medical environment. The article combines **clinical guidelines and expert knowledge** with a data-driven RL model.

CO4 connection: Applying machine learning for real-world decision-making.

This approach can improve safety and ensure that recommendations remain within clinically meaningful choices.

4(b) Proposed Extension
Proposed Experiment: External Validation Using a Different Hospital Dataset

I would extend the research by testing the trained treatment recommendation model on an **independent hospital or healthcare dataset**.

Procedure
Original Training Data
↓
Train Deep RL Model
↓
Generate Treatment Policy
↓
Independent Hospital Dataset
↓
Evaluate Treatment Recommendations
↓
Compare Clinical Outcomes

Why this is feasible

The original study already demonstrates the approach using a large longitudinal diabetes registry. The next step would be to determine whether the learned policy generalizes to patients from a different healthcare environment.

Expected Impact

This experiment would help determine:

- Whether the model generalizes to new populations.
- Whether performance remains stable across hospitals.
- Whether demographic differences affect recommendations.
- Whether the system is suitable for wider clinical deployment.

Final Conclusion

The reviewed article demonstrates how Deep Reinforcement Learning can be applied to personalized Type 2 Diabetes treatment recommendation**. The researchers used a large Singapore Health Services** Diabetes Registry containing **189,520 patients and 6,407,958 outpatient visits from 2013–2018**. They developed three treatment recommendation models covering antiglycemic, antihypertensive, and lipid-lowering treatments. The approach combined **clinical knowledge with data-driven Deep Reinforcement Learning**, making it highly relevant to CO4's reinforcement-learning and decision-making concepts.

The main lesson is that RL can handle **sequential healthcare decisions**, but real-world deployment requires careful validation, privacy protection, explainability, bias assessment, and qualified human oversight. This makes the approach promising for clinical decision support, while additional prospective and external validation would be necessary before relying on it in real patient care.

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand